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A Supersymmetric Flavor Clockwork

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Introduction to Clockwork

Flavor Physics in the quark sector is heavily guided by the Yukawa interactions of the Standard Model (SM),

$$\mathcal{L}_{Yukawa} \supset Y_{ij}^d \bar{q}_L^i H d_R^j + Y_{ij}^u \bar{q}_L^i \tilde{H} u_R^j. \quad (1)$$

These terms are responsible for the quark masses we see in the SM, after the Electro-Weak symmetry breaking occurs.

- Yukawa Couplings: Increasing per generation - **Hierarchical**
- Yukawa Couplings: $\mathcal{O}(1)$ down to $\mathcal{O}(10^{-6})$
- One Higgs vev $\rightarrow$ range of masses over 6 orders of magnitude!
- Why? $\in$ SM Flavor Puzzle

Fermion Mass Hierarchy

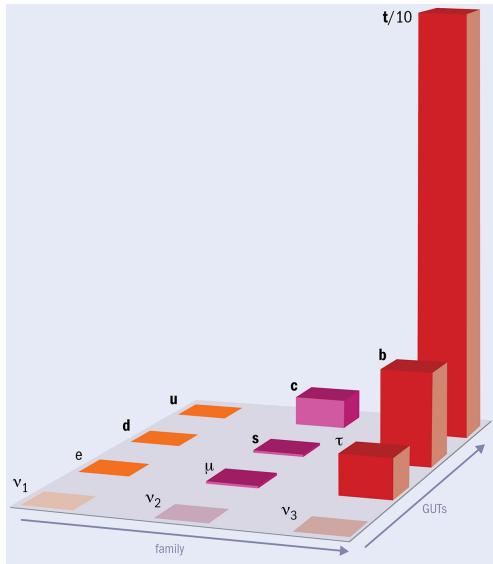


Figure 1: <https://cerncourier.com/a/who-ordered-all-of-that/>

Clockwork: The Setup

The Clockwork mechanism [Giudice et. al [1610.07962](#)] can generate small (or large) scales from $\mathcal{O}(1)$ numbers. As a quick introduction, let us consider the simplest SM Clockwork, for the RH up quark

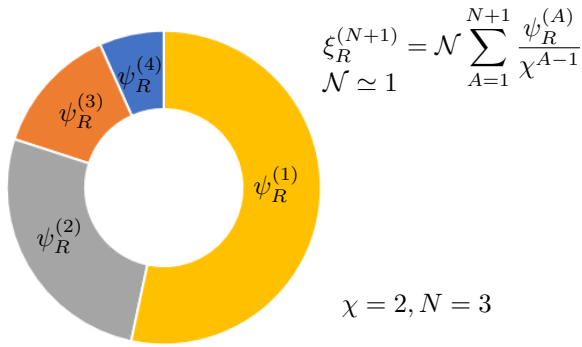
$$u_R \sim (3, 1, \frac{2}{3}):$$

- Introduce N massless LH fermions $\{\psi_L^{(A)}\}$ and $N + 1$ RH counterparts $\{\psi_R^{(A)}\}$
- Global chiral symmetry present; Abelian factors $U(1)_L^{(A)}$ and $U(1)_R^{(A)}$
- Break the symmetry: Spurions $m^{(A)} = m$ charged $(+1, -1)$ under $U(1)_L^{(A)} \times U(1)_R^{(A)}$ and $m_\chi^{(A)} = -m\chi$, charged $(+1, -1)$ under $U(1)_L^{(A)} \times U(1)_R^{(A+1)}$
- One unbroken $U(1)$ factor $\implies$ One massless mode

$$\mathcal{L}_{\text{breaking}} = -m \sum_{A=1}^N \left(\bar{\psi}_L^{(A)} \psi_R^{(A)} - \chi \bar{\psi}_L^{(A)} \psi_R^{(A+1)} + \text{h.c.} \right). \quad (2)$$

Clockwork: The Execution

The exponential suppression is made apparent by going to the mass eigenbasis $\{\xi_R^{(A)}\}$. Observe the massless mode:



Massless mode has an exponentially suppressed overlap with $\psi_R^{(N+1)}$. Identify it with u_R , and make everything else **heavy** ($m \gg v$)! Then,

$$\mathcal{L} \supset -\tilde{Y}^u \bar{q}_L \tilde{H} \psi_R^{(N+1)} \supset -\frac{1}{\chi^N} \tilde{Y}^u \bar{q}_L \tilde{H} u_R \quad (3)$$

We consider the simplest setup:

- All quarks are identified with massless-modes after CW breaking
- Universal Clockwork breaking masses
- Universal gear ratios $\chi^{-1} = \lambda \sim 0.23$
- Gear numbers tune the scales (cf. [Alonso et. al [1807.09792](#)])

$$Y_x^{ik} \simeq \tilde{Y}_x^{ik} \chi^{-N_q^i - N_x^k}, \quad x \in \{u, d\} \quad (4)$$

$$N_q = \begin{pmatrix} 3 \\ 2 \\ 0 \end{pmatrix}, \quad N_u = \begin{pmatrix} 5 \\ 2 \\ 0 \end{pmatrix}, \quad N_d = \begin{pmatrix} 4 \\ 3 \\ 3 \end{pmatrix} \quad (5)$$

SUSY



<https://www.susymagazine.org/article/the-status-of-supersymmetry>

Fixing A Massive Problem

The Higgs mass dependence on the UV cutoff and heaviest fields in the theory make the SM Higgs mass unlikely in its deviation from the Planck mass

Supersymmetry is a well known, elegant, solution to the problem → Provides a superpartner contribution to cancel the aforementioned corrections A Clockwork theory can be extended with the SUSY principle! Our exploration regarded a simple SM flavor extension of the massless quarks and **heavy** gears:

$$W \supset \Phi_{Q_i} \mathbf{m}_{Q_i} \Phi_{Q_i^c} + \Phi_{U_i} \mathbf{m}_{U_i} \Phi_{U_i^c} + \Phi_{D_i} \mathbf{m}_{D_i} \Phi_{D_i^c} \quad (6) \\ + \tilde{Y}_u^{ij} H_u \Phi_{Q_i} \Delta_u^{ij} \Phi_{U_j^c} + \tilde{Y}_d^{ij} H_d \Phi_{Q_i} \Delta_d^{ij} \Phi_{D_j^c} ,$$

where Φ_{ψ_i} , the chiral super-field corresponding to the SM ψ_i , is a vector in Clockwork space, while Δ_x^{ij} is a projection operator, selecting the $(N_q^i + 1, N_x^j + 1)$ gear fields.

SUSY has not been observed, to date. As such, it must be broken at a scale higher than we can currently probe

- Gravity mediated soft SUSY breaking presents us with several new SUSY violating terms that induce squark and gluon mixing
- We consider soft squark masses generated at the SUSY breaking scale $\mathcal{O}(\Lambda_{SUSY})$
- Gaugino masses are independent parameters

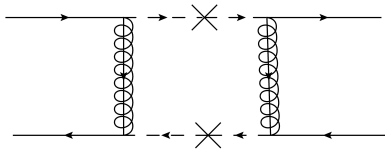
$$\begin{aligned}\mathcal{L}_{soft} \supset & \tilde{\mathbf{Q}}_i^\dagger \mu_{Q_i}^2 \tilde{\mathbf{Q}}_i + \tilde{\mathbf{Q}}_i^{c\dagger} \mu_{Q_i^c}^2 \tilde{\mathbf{Q}}_i^c + (M_Q^2)_{ij} \tilde{\mathbf{Q}}_i^\dagger \Delta_Q^{ij} \tilde{\mathbf{Q}}_j \\ & + \tilde{\mathbf{U}}_i^\dagger \mu_{U_i}^2 \tilde{\mathbf{U}}_i + \tilde{\mathbf{U}}_i^{c\dagger} \mu_{U_i^c}^2 \tilde{\mathbf{U}}_i^c + (M_U^2)_{ij} \tilde{\mathbf{U}}_i^\dagger \Delta_U^{ij} \tilde{\mathbf{U}}_j^c \\ & + \tilde{\mathbf{D}}_i^\dagger \mu_{D_i}^2 \tilde{\mathbf{D}}_i + \tilde{\mathbf{D}}_i^{c\dagger} \mu_{D_i^c}^2 \tilde{\mathbf{D}}_i^c + (M_D^2)_{ij} \tilde{\mathbf{D}}_i^\dagger \Delta_D^{ij} \tilde{\mathbf{D}}_j^c ,\end{aligned}$$

For a simplistic viability test, all soft breaking squark masses are set to m_S , and the gluino mass to m_g , with $\Lambda_{SUSY} \sim \max(m_S, m_g)$.

$$m_{\tilde{q}}^2 \simeq m_S^2 \begin{pmatrix} 1 & \chi^{-1} & \chi^{-3} \\ \chi^{-1} & 1 & \chi^{-2} \\ \chi^{-3} & \chi^{-2} & 1 \end{pmatrix}, \quad (7)$$

$$m_{\tilde{u}}^2 \simeq m_S^2 \begin{pmatrix} 1 & \chi^{-1} & \chi^{-3} \\ \chi^{-1} & 1 & \chi^{-2} \\ \chi^{-3} & \chi^{-2} & 1 \end{pmatrix}, \quad m_{\tilde{d}}^2 \simeq m_S^2 \begin{pmatrix} 1 & \chi^{-1} & \chi^{-1} \\ \chi^{-1} & 1 & 1 \\ \chi^{-1} & 1 & 1 \end{pmatrix} \quad (8)$$

Validity



The tightest bounds on this theory arise from Kaon mixing and decay processes, via squark-gluino mediated box diagrams.

- Regimes of interest: $\Delta m_K^{SUSY} < \delta(\Delta m_K^{SM})$, $|\epsilon_K^{SUSY}| < \delta\epsilon_K^{SM}$
-

$$\langle K^0 | \bar{d}_L^\alpha \gamma_\mu s_L^\alpha \bar{d}_L^\beta \gamma^\mu s_L^\beta | \bar{K}^0 \rangle = \frac{1}{3} B_1(\mu) m_K f_K^2,$$

$$\langle K^0 | \bar{d}_R^\alpha s_L^\alpha \bar{d}_L^\beta s_R^\beta | \bar{K}^0 \rangle = \frac{1}{4} B_4(\mu) m_K f_K^2 \frac{m_K^2}{(m_d(\mu) + m_s(\mu))^2},$$

$$\langle K^0 | \bar{d}_R^\alpha s_L^\beta \bar{d}_L^\beta s_R^\alpha | \bar{K}^0 \rangle = \frac{1}{12} B_5(\mu) m_K f_K^2 \frac{m_K^2}{(m_d(\mu) + m_s(\mu))^2}$$

$$C_i^{SUSY} \propto \alpha_s^2 / \Lambda_{SUSY}^2$$

- $B_i(3\text{GeV})$ from lattice computations [Carrasco et. al [1505.06639](#)]
- Use RGE matrices to flow Wilson coefficients $C_i^{SUSY}(\alpha_s; m_S, m_g)$ from $\Lambda_{SUSY} \rightarrow 3\text{ GeV}$, to evaluate $\vec{C}_{SUSY} \cdot \langle \vec{O} \rangle$
- The real part is Δm_K^{SUSY} , and the imaginary, ϵ_K^{SUSY}

Valid Parameter Space

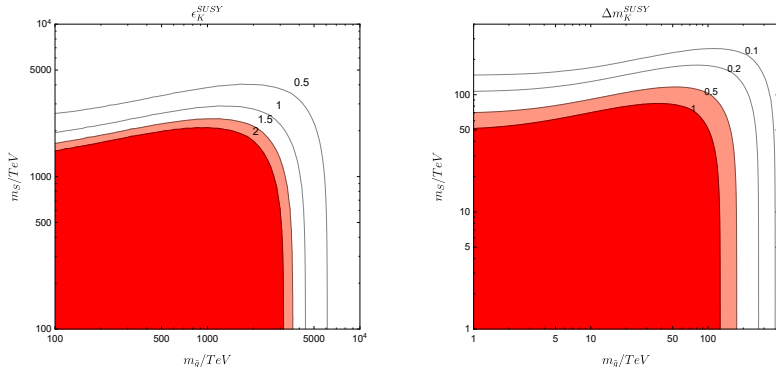


Figure 2: The ϵ_K plot indicates some contours labelling the ratio of the Clockwork SUSY contribution to ϵ_K to the theoretical uncertainty of the SM ϵ_K prediction in [Brod et. al, [1911.06822](#)]. The Δm_K plot tracks the ratio of the Clockwork SUSY contribution to Δm_K to the Standard Model experimental value found in the [PDG review](#).

Conclusion

- Clockwork is an answer to a part of the SM Flavor Puzzle: **Hierarchies** in quark masses, and the CKM matrix
- It is compatible with SUSY, and can thus address another big discrepancy in scales: m_{higgs}, M_P
- The full theory is valid if $\Lambda_{SUSY} > 3 \text{ PeV}$
- The "simplest" model works, but must be fine tuned to ameliorate the Higgs sensitivity to the new UV of $\Lambda_{SUSY} > 3 \text{ PeV}$
- Further flavor mixing and fine-tuning can be incorporated and tweaked by having non-universal masses and gear ratios

The $\tan \beta$ Suppression (Additional Considerations)

While considering a SUSY BSM theory, recall that we necessarily speak of multiple Higgs fields - H_u and H_d in this theory

Consider $\tan \beta = \langle H_u \rangle / \langle H_d \rangle$, where the region of validity is approximately $1 < \tan \beta < 65$, from constraints on the evolution of Yukawa couplings and LFV considerations

Large $\tan \beta$ = down type suppression. Let $n = \lfloor \log_\chi \tan \beta \rfloor$:

$$N_d = \begin{pmatrix} 4 - n \\ 3 - n \\ 3 - n \end{pmatrix} \quad (9)$$

Asymptotic Freedom and Landau Poles (Additional Considerations)

The asymptotic freedom of QCD is realized only in a theory with a few flavors of quarks. This is called into question above the Clockwork scale (where the first heavy gears become dynamic)

The number of gears makes it inevitable to develop a Landau Pole. This is not an issue as long as this occurs at an extremely high scale ($> M_P$), which is determined by the Clockwork scale, Λ_{CW}

$$\Lambda_{LP} = \Lambda_{CW} \exp \left[\frac{2\pi}{(24 - 3n)\alpha_s(\Lambda_{CW})} \right], \quad (10)$$

From this, we obtain a lower bound on $\Lambda_{CW} > 10^{16}$! Visually,

Asymptotic Freedom and Landau Poles (Additional Considerations)

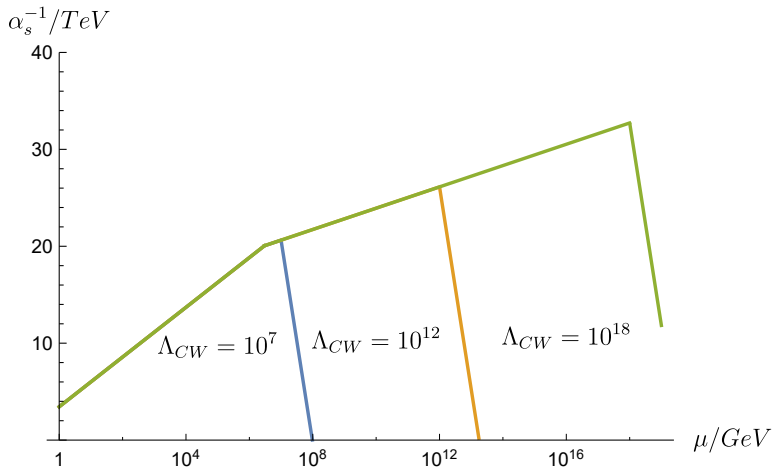


Figure 3: Running of α_s , and Landau Poles at various scales.

Questions?